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Robotics & Perception Engineer · **SLAM, 3D Reconstruction & Gaussian Splatting** · **Edge-to-Cloud ML Systems**

SUMMARY

Robotics and computer vision engineer who builds perception systems end to end, from raw sensor data through deployed inference. Five-plus years of production computer vision and robotics work, most of it inside startups, spans SLAM, 3D reconstruction, and edge-deployed deep learning on real hardware. Currently pursuing an M.S. in Robotics, researching Gaussian Splatting, multi-sensor SLAM capture, and simulation-based autonomy stacks.

EDUCATION

M.S. Robotics — **Aug 2024 – Jan 2027 (Expected)**
University of Minnesota – Twin Cities — Minneapolis, MN
Coursework: Computer Vision, Intelligent Robotic Systems, Image Processing, Deep Learning, NLP.

B.Tech, Electronics & Communication Engineering — **Jun 2017 – Jun 2021**
GITAM University — Hyderabad, India

TECHNICAL SKILLS

Languages: Python, C++, CUDA, Go, C, Rust

Build Systems: CMake, Bazel, building large C++/Python projects from source with custom patches (e.g. PyTorch)

SLAM & 3D Perception: cuVSLAM, ORB-SLAM3, RTAB-Map, KISS-ICP, visual-inertial SLAM, sensor fusion, Kalman filters, 3D reconstruction, multi-view geometry, coordinate-frame transforms (SO(3)/SE(3), quaternions, SLERP)

Sensors & Calibration: RealSense D435i (RGB-D + IMU), LiDAR, LiDAR-camera extrinsics, stereo rectification, camera intrinsics/extrinsics, sensor time synchronization

Robotics Middleware & Simulation: ROS/ROS2, FastDDS, Isaac Sim (Omniverse), Gazebo, ArduPilot/PX4, MAVLink

Machine Learning & Vision-Language: PyTorch, TensorFlow, CNNs, ViT, Transformers, CLIP, DINOv2, Gaussian Splatting, NeRF

Edge & On-Device Deployment: Jetson Orin, Jetson AGX, Raspberry Pi, TensorRT, ONNX, quantization, pruning, distillation

Infrastructure & Cloud: Kubernetes, Triton Inference Server, KServe, KNative, Docker, gRPC, Argo Workflows, AWS/GCP/Azure

Data & Observability: MongoDB, PostgreSQL, Redis, Kafka, RabbitMQ, Prometheus, Grafana, Foxglove

EXPERIENCE

Graduate Research Assistant — **Sep 2025 – Present**
University of Minnesota – Twin Cities — Minneapolis, MN

- Built an end-to-end ROS 2 mapping and navigation pipeline for outdoor robots: an Intel RealSense D435i feeding NVIDIA Isaac ROS cuVSLAM 3.2 for real-time visual-inertial SLAM, with a synchronized 2 Hz capture node defining the data-collection protocol for downstream training.
- Built a custom Gaussian Splatting pipeline from scratch for high-fidelity 3D scene reconstruction, and developed feed-forward Gaussian Splatting methods that work from minimal input views.
- Audited the RealSense driver configuration end to end and found 11 distinct misconfigurations silently degrading SLAM tracking and data quality: an invalid accelerometer rate, sync-induced frame jitter, IMU timestamps racing ahead of images. Fixed each to root cause and tightened jitter tolerance from 34 ms to 19 ms.
- Researched and implemented city-scale Gaussian Splatting with memory-efficient scene representations for large outdoor environments. The pipeline runs on a single GPU with under 5 GB of VRAM, not a multi-GPU cluster.
- Wrote a three-way coordinate-frame translation layer across ROS, OpenGL, and COLMAP conventions, plus a quaternion-SLERP correction scheme for loop-closure pose jumps, keeping poses geometrically consistent for both training and deployment.
- Fused CLIP and DINOv2 features into the splat representation, so a reconstructed environment can be queried and localized in natural language, an early step toward language-grounded spatial reasoning.
- Designed a 3-machine distributed ROS 2 system, Dockerized on host networking over a Tailscale mesh: chose a FastDDS Discovery Server over CycloneDDS because multicast does not traverse VPN tunnels, then bisected and fixed a Tailscale `tun0` userspace-networking bug via FastDDS interface whitelisting.

- Architected a full sim-in-the-loop autonomy stack on NVIDIA Isaac Sim, chaining visual-inertial SLAM, 3D reconstruction, navigation, and autonomous exploration into one GPU-accelerated pipeline for a ground robot. Ported an abandoned robot demo across a major platform rewrite to get it running, then designed the custom multi-sensor rig and every stage built on top of it.
- Migrated an abandoned robot simulation off a discontinued NVIDIA Isaac Sim release onto the current one, fixing nine categories of breaking changes: a full Python API namespace migration, a URDF importer rewritten from scratch against the new class-based API, and changed OmniGraph node schemas. Validated every fix by running the robot end to end rather than trusting documentation.
- Diagnosed a silent multi-camera sensor-fusion failure by proving two cameras sixty degrees apart could not triangulate depth, then redesigned the camera rig to fix it and verified the result against live inertial ground-truth measurements. Built a measurement tool to derive a lidar sensor's intrinsic parameters directly from live data, since the values could not be computed from its configuration alone.
- Cut camera data out of the pipeline's slower communication path by enabling zero-copy GPU image transport between the perception and navigation stages, and solved the container-level permission requirements the change introduced. Also retuned the simulation's real-time data rates and cut the lidar's data volume four-fold after measuring it was starving the navigation stack of fresh sensor data.
- Authored a custom multi-sensor payload from scratch in URDF/xacro (dual RGB-D cameras plus a 3D spinning lidar) and built the matching Isaac Sim sensor publishers for RGB, depth, stereo infrared, and IMU streams, keeping every simulated topic consistent with the real sensor driver's layout so the same downstream code runs against either.
- Decoupled the simulator's physics tick rate from its render rate, running the physics engine at 180 Hz against 60 Hz rendering, after measuring that render-rate-only updates left the inertial data too sparse for the visual-inertial SLAM stack to track reliably.

Robotics Perception Intern **Jun 2025 – Dec 2025**
Parallax Worlds (Startup) San Francisco, CA

- Managed Kubernetes cluster state and infrastructure provisioning with Terraform, and set up the deployment workflow end to end on ArgoCD and Argo Workflows, giving the team a GitOps-driven path from a config change to a running cluster.
- Deployed the Gaussian Splatting reconstruction pipeline on Kubernetes with dynamic GPU allocation across H100, A100, and T4 tiers, cutting processing time 30%.
- Built a physics-estimation pipeline that infers Young's modulus, Poisson's ratio, density, and friction for reconstructed deformable and rigid objects from visual features and texture, connecting 3D reconstruction to physical-world understanding.
- Built the monitoring and alerting layer on Prometheus and Grafana so failures surfaced instead of hiding in logs, cutting failure response time 40%.

Senior Computer Vision Scientist, Core Team **Jan 2019 – Jul 2024**
Upjao Agrotech (Startup) Ahmedabad, India

- Built detection, segmentation, and classification models (ViT, Faster-RCNN, Mask-RCNN, YOLO) with custom loss and activation functions, gaining up to 20% accuracy.
- Deployed vision models to Jetson and Raspberry Pi boards under hard latency budgets using quantization, pruning, and distillation: 40% faster inference, 20% smaller models.
- Designed a serverless inference platform on Kubernetes using Triton Inference Server, KServe, KNative, and gRPC, serving multiple models under 500 ms latency.
- Built streaming and bidirectional gRPC servers on top of NVIDIA Triton Inference Server, plus custom extension interfaces for specialized inference workloads. Connected these servers to Android and JavaScript client apps using gRPC, ConnectRPC, and REST, including Kotlin gRPC clients for the Android integration.
- Improved inference performance 50% and cut infrastructure costs 60% through multithreading, vectorization, and FP16/INT8 precision on A100 and T4 GPUs.
- Built a model-caching algorithm that reduced scale-up time from 15–20 minutes to 3–4 seconds.
- Led and mentored a team of 6+ engineers, and secured roughly $200K in NVIDIA and Azure cloud-credit partnerships.
- Presented technical updates and new AI approaches directly to investors, company directors, and other non-technical stakeholders, and worked daily across sales, testing, and engineering teams to translate what a model could and could not do into decisions the wider team could act on.

AI / ML Developer **Jun 2021 – Apr 2023**
Tata Consultancy Services Hyderabad, India

- Selected into TCS Rapid Labs, an internal AI and robotics R&D cohort operating on fast three-month proof-of-concept cycles. Won the Best Team Award.
- Led a team of 5 engineers building full-stack proof-of-concepts, and architected the cloud and HPC infrastructure behind them.
- Spearheaded an in-house serverless inference and deployment platform for computer vision, NLP, and multi-modal models.

– Built a data-cleaning algorithm to detect and remove duplicate data, gaining 20% efficiency.

SELECTED PROJECTS

– **Real-Time Multi-Sensor 3D Mapping** ROS 2, C++, Python, Jetson Orin, LiDAR, RealSense RGB-D — Outdoor ground-robot SLAM pipeline on a Jetson Orin, fusing LiDAR, RealSense RGB-D, and IMU for autonomous agricultural field mapping. Solved LiDAR-camera extrinsic calibration with the Koide direct visual-LiDAR method, tuned KISS-ICP odometry for stable outdoor trajectories, and ran Isaac ROS Visual SLAM alongside a distributed multi-machine ROS 2 graph discovered over a Tailscale-meshed FastDDS server. Keyframe output feeds directly into COLMAP and Gaussian Splatting reconstruction downstream.

– **Autonomous Drone Navigation with Stereo SLAM** C++, ROS 2, ArduPilot, Jetson Orin Nano, Pixhawk — Solo build of a GPS-denied navigation stack on a custom quadcopter: a stereo rig from two Raspberry Pi cameras, hand-calibrated, feeding ORB-SLAM3 and RTAB-Map for real-time visual-inertial localization and mapping. Ran on a Jetson Orin Nano paired with a Pixhawk flight controller over ROS 2, validated end to end in simulation, with fallback logic in place for SLAM tracking loss, before flight testing.

– **Indoor Mapping with Semantic 3D SLAM** ROS 2, RTAB-Map, ORB-SLAM3, TurtleBot 3, Isaac Sim — Ground-robot SLAM on a TurtleBot 3, fusing RTAB-Map and ORB-SLAM3 with instance segmentation to produce semantically labeled 3D maps in GPS-denied indoor environments, validated first in Isaac Sim. Reached 95% mapping and localization accuracy and a 70% improvement in navigation efficiency over the unfused baseline, with dynamic obstacle avoidance layered on top of the landmark-tracked map.

– **Image-Based Visual Servoing for Robotic Arm Control** TensorRT, YOLOv8, Custom Stereo, Open3D — Closed a real-time perception-to-control loop on a robotic arm: a custom stereo camera and the "Stereo Anything" depth model generate 3D point clouds from disparity maps, while YOLOv8 segmentation identifies targets. Optimized both models with TensorRT down to sub-5 ms inference, budgeting perception-control loop latency carefully to keep the closed-loop visual-servoing control law stable.

– **RL-Based Robotic Manipulation with Koopman Operator** Isaac Sim, PPO, Koopman Operator, PyTorch — Trained a robotic picking policy in Isaac Sim with PPO, layering a Koopman-operator linear model on top to approximate the arm's non-linear dynamics for smoother control. Designed the reward function and sim-to-real transfer path from scratch, cutting task completion time 30%, collision errors 25%, and erratic motion 40% against the un-linearized baseline.

– **Manipulator Simulation & Control Pipeline** MuJoCo, Python, Jacobian IK — Built a full sim-to-control pipeline from scratch in MuJoCo for a 6-DOF SO-101 arm: diagnostic tooling to validate a community MJCF conversion against the model's joint/actuator/body topology, a real-time-synced viewer for visual verification, a joint-by-joint sinusoidal sweep to sanity-check actuator wiring and limits, and a Jacobian-based damped least-squares inverse-kinematics solver ($\Delta q = J^T(JJ^T + \lambda^2 I)^{-1}\Delta x$) implemented from first principles to drive the end effector along a circular trajectory.

– **Isaac Lab Manipulation Environment** NVIDIA Isaac Lab, Isaac Sim, Gym — Built a manager-based Gym task package in Isaac Lab for the same SO-101 arm, registering three training and teleoperation environments and implementing both a 7D IK-delta-plus-gripper and a 9D joint-delta teleoperation action space through `isaaclab.devices`. Diagnosed and fixed an asset-mounting bug end to end (a stale hardcoded table-height assumption compounded by an incorrect quaternion component ordering) by measuring the table's true top surface at runtime via its bounding-box cache, while working through Isaac Lab 3.0's beta-cycle breaking changes as an early adopter, including a platform-wide quaternion convention flip from WXYZ to XYZW.

– **Custom Quadcopter Design & Development** Pixhawk, F450 Frame, PX4, MAVLink — End-to-end mechatronic build: component selection through battery and propeller trade-off analysis, long-range telemetry configuration, and MAVLink-level debugging to enable advanced autonomous flight modes. Reached 2.8 km of constant-telemetry range, 20 minutes of flight time, and a 6 kg payload capacity.

– **Image Super Sampling (SRGAN)** PyTorch, CUDA, GAN — Real-time 720p-to-4K video upscaling with an SRGAN architecture, stabilized with a perceptual loss term to avoid the training collapse GANs are prone to. Profiled GPU memory usage and added dynamic memory allocation for large video buffers, cutting memory use 40% while explicitly reasoning about the speed-versus-quality trade-off in the super-resolution pipeline.

– **Real-Time Audio Noise Cancellation** Wave-UNet, STFT, Torchaudio — Wave-UNet architecture for real-time noise cancellation and speech enhancement, trained on Mozilla Common Voice and UrbanSound8k with an optimized STFT/iSTFT pipeline and one-cycle learning-rate scheduling, reaching an MSE of approximately 0.2. The results were later published in *Design Engineering* (2021; see Publications below).

– **Pothole Detection** ResNet50, RetinaNet, Albumentations — Production-grade pothole detector built on a RetinaNet head over a ResNet50 backbone, with a semi-synthetic data generation pipeline and a heavy Albumentations augmentation stack to compensate for limited labeled data, trained and evaluated against PascalVOC/COCO-format annotations. Reached mAP above 0.85 at 0.95 IoU.

PUBLICATIONS, PATENTS & RECOGNITION

– **Publication:** Kumar, Ch Praveen, K. Praveen Kumar, KSS Vamsi Krishna, and T. Shwetha. "Noise Cancellation and Speech Enhancement Using Multi-Layer Convolutional Neural Network." *Design Engineering* (2021): 921–926.

– **Patents (granted):** System for Quality Assessment of Agricultural Product (436541); A Method and System for Encoding and Decoding Data Using Markers (420903).

– **Patents (provisional & filed):** An Adjustable Apparatus for Uniform Image Acquisition (provisional, 202221022993); Generic Classification Using Very Less Data (filed); Cattle Identification Using Face Recognition (filed).
– **Recognition:** Best Team Award, TCS Rapid Labs Cohort (2023); Juror, LaElevitia Hackathon, evaluated projects from 80+ students (2023); Organizer & Mentor, Innovation Centre Club, AI and robotics outreach and workshops (2018–2021).

CERTIFICATIONS

– **Deep Learning Specialization**, DeepLearning.AI (Coursera, 2020): Convolutional Neural Networks; Sequence Models; Improving Deep Neural Networks (Hyperparameter Tuning, Regularization, Optimization); Structuring Machine Learning Projects.